

Daniel Gonzalez Cedre

cedre.tech · cedre@illinois.edu

Experience

Teaching Assistant Professor

SIEBEL SCHOOL OF COMPUTING AND DATA SCIENCE

University of Illinois Urbana-Champaign

AUG 2025 – PRESENT

Education

Doctor of Philosophy · Computer Science

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

University of Notre Dame

JUN 2019 – JUL 2025

Master of Science · Computer Science

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

University of Notre Dame

JUN 2019 – APR 2023

Master of Science · Financial Mathematics

DEPARTMENT OF MATHEMATICS

Florida State University

AUG 2016 – MAY 2019

Bachelor of Science · Mathematics · Computer Science · *cum laude*

DEPARTMENT OF MATHEMATICS AND STATISTICS · SCHOOL OF COMPUTING AND INFORMATION SCIENCES

Florida Int'l University

AUG 2012 – MAY 2016

Associate of Arts · Mathematics · *magna cum laude*

SCHOOL FOR ADVANCED STUDIES, WOLFSON CAMPUS

Miami-Dade College

JUN 2010 – APR 2012

Awards & Honors

2024	Outstanding Graduate Student Teaching Award	Kaneb Center for Teaching & the Graduate School	University of Notre Dame
2024	Kaneb Outstanding Instructor of Record	Department of Computer Science and Engineering	University of Notre Dame
2024	CSE Outstanding TA Award	Department of Computer Science and Engineering	University of Notre Dame
2024	Outstanding Instructor Honorable Mention	Graduate Student Government	University of Notre Dame
2019	Deans' Graduate Fellowship	The Graduate School	University of Notre Dame
2017	Dean's Scholarship	The Graduate School	Florida State University
2016	GEM University Fellow	National GEM Consortium	University of Chicago
2016	Outstanding Achievement in Mathematics	College of Arts, Sciences, and Education	Florida Int'l University
2015	McNair Scholar, 12 th cohort	McNair Scholars Program	Florida Int'l University
2015	Second place for "Monotone Catenary Degree [...]"	McNair Scholars Research Conference	Florida Int'l University
2014	Third place for "Pancake Simulator"	HackFSU Hack-a-thon	Florida State University
2012	National Hispanic Scholarship	Office of Admissions	Florida Int'l University
2012	Florida Bright Futures Scholarship	Office of Student Scholarship and Grants	State of Florida

Research & Publications

2024	This Probably Looks <i>Exactly</i> Like That: An Invertible Prototypical Neural Network	ECCV
	Daniel Gonzalez Cedre* · Zachariah Carmichael* · Timothy Redgrave* · Walter Scheirer	*EQUAL CONTRIBUTION
2023	Dynamic Vertex Replacement Grammars	arXiv
	Daniel Gonzalez Cedre · Justus Isaiah Hibshman · Timothy La Fond · Grant Boquet · Tim Weninger	
2023	Motif Mining: Finding and Summarizing Remixed Image Content	WACV
	William Theisen · Daniel Gonzalez Cedre · Zachariah Carmichael · Daniel Moreira · Tim Weninger · Walter Scheirer	
2022	The Infinity Mirror Test for Graph Models	TKDE
	Satyaki Sikdar · Daniel Gonzalez Cedre · Trenton W. Ford · Tim Weninger	
2021	Temporal Egonet Subgraph Transitions	arXiv
	Daniel Gonzalez Cedre · Sophia Abraham · Lucas Parzianello · Eric Tsai	
2021	Joint Subgraph-to-Subgraph Transitions	WSDM
	Justus Hibshman · Daniel Gonzalez Cedre* · Satyaki Sikdar* · Tim Weninger	*EQUAL CONTRIBUTION
2015	Monotone Catenary Degree in Numerical Monoids	arXiv
	Daniel Gonzalez Cedre · Cameron Wright · Jenna Zomback	

Talks & Presentations

2025	Rules and Distributions for Explainable Machine Learning	PhD Thesis Defense	University of Notre Dame
2024	This Probably Looks <i>Exactly</i> Like That	Poster Presentation	European Conference on Computer Vision
2024	Undergraduate Engineering Discernment Lecture	Invited Guest Lecture	University of Notre Dame
2023	Explaining Anomalies in Graphs with Grammars	Internship Talk	Deloitte
2023	A Transformational Approach to Graph Learning	PhD Oral Candidacy	University of Notre Dame
2023	Undergraduate Engineering Discernment Lecture	Invited Guest Lecture	University of Notre Dame
2022	Undergraduate Engineering Discernment Lecture	Invited Guest Lecture	University of Notre Dame
2021	Mining Temporal Hypergraphs with Graph Grammars	Invited Guest Lecture	Rose-Hulman Institute of Technology
2020	Undergraduate Engineering Discernment Lecture	Invited Guest Lecture	University of Notre Dame
2020	The Infinity Mirror Test for Graph Generators	Full Talk	SIAM Network Science
2020	The Infinity Mirror Test for Graph Generators	Poster Presentation	ND CSE 14 th Annual Poster Conference
2015	Monotone Catenary Degree in Numerical Monoids	Poster Presentation	FIU McNair Scholars Research Conference

Teaching

Discrete Structures · CS 173	University of Illinois Urbana-Champaign
INSTRUCTOR OF RECORD · TBD	FALL 2026
Discrete Structures · CS 173	University of Illinois Urbana-Champaign
INSTRUCTOR OF RECORD · 145 STUDENTS	SPRING 2026
Modeling and Learning in Data Science · CS 307	University of Illinois Urbana-Champaign
INSTRUCTOR OF RECORD · 200 STUDENTS	SPRING 2026
Discrete Structures · CS 173	University of Illinois Urbana-Champaign
INSTRUCTOR OF RECORD · 570 STUDENTS	FALL 2025
Warrior Scholars Program · OUTREACH PROGRAM	University of Notre Dame
INSTRUCTOR AND MENTOR · 6 STUDENTS	SUMMER 2025
Principles of Computing · CSE 10001	University of Notre Dame
INSTRUCTOR OF RECORD · 36 STUDENTS	FALL 2024
Discrete Mathematics · CSE 20110	University of Notre Dame
INSTRUCTOR OF RECORD · 31 STUDENTS	SPRING 2024
Discrete Mathematics · CSE 20110	University of Notre Dame
INSTRUCTOR OF RECORD · 180 STUDENTS	FALL 2023
CSE Summer Enrichment Program · LECTURE SERIES	University of Notre Dame
INSTRUCTOR AND COACH · 20 STUDENTS	SUMMER 2023
Discrete Mathematics · CSE 20110	University of Notre Dame
INSTRUCTOR OF RECORD · 21 STUDENTS	SPRING 2023
Graph Theory · DIRECTED READING	University of Notre Dame
CO-ORGANIZER · 1 STUDENT	FALL 2022
Discrete Mathematics · CSE 20110	University of Notre Dame
INSTRUCTOR OF RECORD · 25 STUDENTS	SPRING 2022
Discrete Math I · MAD 2104	Florida State University
RECITATION INSTRUCTOR · 60 STUDENTS	SPRING 2019
Precalculus Algebra · MAD 1104	Florida State University
INSTRUCTOR OF RECORD · 35 STUDENTS	FALL 2018

Mentorship

2025	Warrior Scholars Program	Project Mentor & Advisor for 6 Students	University of Notre Dame
2023	CSE Summer Enrichment Program	Coach & Instructor for 20 Students	University of Notre Dame

Collaborations

Data Scientist & AI Graph Expert

AI CENTER FOR EXCELLENCE · *Sanmitra Bhattacharya* · *Salvador Aguiñaga*

- Worked to develop an intrinsically-explainable graph neural network based on graph grammars

Intern Research Scientist

APPLIED STATISTICS GROUP · *Grant Boquet* · *Timothy La Fond*

- Developed a dynamic vertex-replacement graph grammar
- Worked to find optimal dendrogram decompositions for vertex-replacement graph grammars
- Modeled temporal graphs with hidden Markov models and vertex-replacement graph grammars

Contactless Fingerprint Collection

COMPUTER VISION RESEARCH LAB · *Adam Czajka*

- Collaborated with Aidan Draper
- Sponsored by West Virginia University

PURE Math Research Program

DEPARTMENT OF MATHEMATICS · *Roberto Pelayo* · *Brian Wissman*

- Studied properties of numerical monoids and their catenary degrees
- Collaborated with Cameron J. Wright and Jenna Zomback

Service

FACULTY VOLUNTEER	<i>University of Illinois Urbana-Champaign</i> · <i>Engineering Open House</i>
COMMITTEE MEMBER	<i>University of Illinois Urbana-Champaign</i> · <i>Academic Appeals Committee</i>
PEER REVIEWER	<i>Springer DMKD</i> · <i>Data Mining and Knowledge Discovery</i>
PEER REVIEWER	<i>Springer JoCO</i> · <i>Journal of Combinatorial Optimization</i>
PEER REVIEWER	<i>ACM WSDM</i> · <i>Web Search and Data Mining</i>
PEER REVIEWER	<i>IEEE TKDE</i> · <i>Transactions on Knowledge, Data, and Engineering</i>
PEER REVIEWER	<i>IEEE LACS</i> · <i>International Conference on Autonomous Systems</i>
STUDENT VOLUNTEER	<i>University of Notre Dame</i> · <i>Graduate Student Recruitment Representative</i>

Deloitte

MAY 2023 – AUG 2023

Lawrence Livermore Nat'l Laboratory

MULTIPLE DATE RANGES

MAY 2022 – AUG 2022

JUN 2021 – AUG 2021

NOV 2020 – FEB 2021

University of Notre Dame

JUN 2019 – JAN 2020

University of Hawai'i at Hilo

JUN 2015 – JUL 2015